

Using the Chain Rule with the Product Rule or the Quotient Rule (Homework)

Find $\frac{dy}{dx}$.

① $y = x^4 \cos(x^3)$

② $y = (3x^4 - 4)^6 x^2$

③ $y = x^3 \sqrt{1 - 4x + 6x^2}$

Find $f'(x)$.

④ $f(x) = \sqrt[5]{x+100} \tan(4x)$

⑤ $f(x) = (x^2+x)^4 \sec \sqrt[3]{x}$

Differentiate.

$$\textcircled{6} \ y = \frac{3x^2}{\csc(x^3) + 4}$$

$$\textcircled{7} \ f(x) = \frac{(x^2 + 9)^2}{4x^{10} - 2}$$

Find y' .

$$\textcircled{8} \ y = \frac{-1}{(x+3)^8}$$

$$\textcircled{9} \ y = \frac{3}{\sin^2 x - 1}$$

$$(10) \ y = \frac{\cot(-2x^6)}{(x^2+7)^9}$$

Homework Answers

$$\textcircled{1} \quad 4x^3 \cos(x^3) - 3x^6 \sin(x^3)$$

$$\textcircled{2} \quad 72x^5(3x^4-4)^5 + 2x(3x^4-4)^6 \quad \text{or} \quad 2x(3x^4-4)^5(39x^4-4)$$

$$\textcircled{3} \quad 3x^2 \sqrt{1-4x+6x^2} + \frac{6x^4-2x^3}{\sqrt{1-4x+6x^2}} \quad \text{or} \quad \frac{24x^4-14x^3+3x^2}{\sqrt{6x^2-4x+1}}$$

$$\textcircled{4} \quad \frac{\tan(4x)}{5\sqrt[5]{(x+100)^4}} + 4\sqrt[5]{x+100} \sec^2(4x) \quad \text{or} \quad \frac{\tan(4x) + 20x \sec^2(4x) + 2000 \sec^2(4x)}{5\sqrt[5]{(x+100)^4}}$$

$$\textcircled{5} \quad 4(x^2+x)^3(2x+1) \sec^3 \sqrt[3]{x} + \frac{(x^2+x)^4 \sec^3 \sqrt[3]{x} \tan \sqrt[3]{x}}{3\sqrt[3]{x^2}} \quad \text{or}$$

$$8x^4(x+1)^3 \sec^3 \sqrt[3]{x} + \frac{1}{3} \sqrt[3]{x^{10}}(x+1)^4 \sec^3 \sqrt[3]{x} \tan \sqrt[3]{x} + 4x^3(x+1)^3 \sec^3 \sqrt[3]{x}$$

$$\textcircled{6} \quad \frac{9x^4 \cot(x^3) \csc(x^3) + 6x \csc(x^3) + 24x}{(\csc(x^3) + 4)^2}$$

$$\textcircled{7} \quad \frac{4x(x^2+9)(4x^{10}-2) - 40x^9(x^2+9)^2}{(4x^{10}-2)^2} \quad \text{or} \quad \frac{-2(x^2+9)(3x^{11}+45x^9+x)}{(2x^{10}-1)^2}$$

$$\textcircled{8} \quad \frac{8}{(x+3)^9}$$

$$\textcircled{9} \quad -\frac{6 \sin x \cos x}{(\sin^2 x - 1)^2}$$

$$\textcircled{10} \quad \frac{12x^5 \csc^2(-2x^6)(x^2+7)^9 - 18x(x^2+7)^8 \cot(-2x^6)}{(x^2+7)^{18}}$$

Factor $(x^2+7)^8$ from the numerator.

$$\frac{12x^5(x^2+7) \csc^2(-2x^6) - 18x \cot(-2x^6)}{(x^2+7)^{10}}$$

or

$$\frac{18x \cot(2x^6) + 12x^7 \csc^2(2x^6) + 84x^5 \csc^2(2x^6)}{(x^2+7)^{10}}$$